Chapter 5 — Solids and Fluid Dynamics

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Stress equals: (a) A/F (b) ΔL/L (c) F × A (d) F/A

2. Strain has: (a) newton (b) metre (c) pascal (d) no unit

3. Young's modulus: (a) ΔL × L (b) stress/strain (c) F × A (d) strain/stress

4. Upthrust equals: (a) mg (b) gV (c) ρgV (d) ρV

5. Continuity equation: (a) $P + \frac{1}{2}\rho v^2 = c$ (b) $F = 6\pi\eta rv$ (c) $A_1v_1 = A_2v_2$ (d) $Y = \sigma/\varepsilon$

6. In horizontal fast flow, pressure: (a) zero (b) high (c) low (d) same

7. Stokes' law: (a) F = ρgV (b) $F = 6\pi\eta rv$ (c) F = kx (d) F = ma

8. At terminal velocity, acceleration: (a) g (b) negative (c) maximum (d) zero

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is strain a dimensionless quantity?

2. Why does a steel ship float while a steel block sinks?

3. Why does squeezing a rubber pipe increase the speed of water?

4. Why is pressure lower where a fluid moves faster?

5. Why is the acceleration zero at terminal velocity?

6. How does the area under a force–extension graph give energy stored?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Derive the equation of continuity and Bernoulli's equation for an ideal fluid. **(4)**

(b) Explain any four applications of Bernoulli's equation. **(4)**

Q.4 (a) Explain the important points on a stress–strain curve for a ductile material. **(4)**

(b) Derive the expression for terminal velocity using Stokes' law. **(4)**

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